

SOCURE GROWTH EXERCISE

/ PIPELINE TOPOGRAPHY

30 mock leads → suppress → enrich → score → route → enable. A **deterministic pipeline** with exactly **two gated language leaves** — the LLM appears only where language is the deliverable or the world is genuinely ambiguous; code decides everything a metric tracks.

DOCUMENT Socure.ARCHITECTURE.md	STAGES 8
LLM TOUCHPOINTS 3 (Claude Haiku)	NEURAL AUTHORITY 0/30 tiers · 0/30 routes

● symbolic · code decides ● neural · LLM touchpoint ● human gate ● invariant held

01 PIPELINE RAIL

Eight stages, run as Trigger.dev tasks. Every stage names its executor, its decision source, and the invariant the run must hold.

1 Ingest

Parse 30 leads; identity fields (name / email / domain / title) are never mutated.

lib/csv.ts

SYMBOLIC

► 30/30 identity intact

2 Gate 0 — Suppress

Registrable-domain match against the 34-customer list. Known customers are never enriched on acquisition spend.

lib/suppression.ts + data/socure-customer-list.csv · decides by set membership

→ **expansion book** — suppressed accounts are routed, never dropped; Part 2 re-points enrich→score→route at this book

SYMBOLIC

- ▶ suppressed → expansion, never dropped

3 Account resolution

Group leads into accounts by registrable domain; `account.score = MAX(leads)` ; ownership-ledger check.

`lib/domain.ts · lib/ownership.ts + memory/account-owners.json`

SYMBOLIC

- ▶ owned accounts keep their owner

4 Enrich — cascade

Waterfall of sources by confidence (§02). The LLM is the residue: classifies, never scores; fires only when every source misses or confidence < 0.5; output clamped to ≤ 0.6 in code.

`connections/* · lib/enrich.ts · config/enrichment.json`

SYMBOLIC

+ BOUNDED NEURAL

- ▶ every field carries source + confidence – `assertEnrichmentContract`

5 Score

4-dimension rubric, 0–100: hot ≥ 75 · warm 50–74 · cold < 50. Vertical matches domain evidence first — an LLM label never overrides a domain keyword.

`lib/scoring.ts · config/icp-rubric.json` (hard rule: feeds tracked metrics)

SYMBOLIC

- ▶ every score ships breakdown · every DQ ships `dq_reason`

6 Route

Owned → owner, never reassigned. Unowned → vertical→rep rules; tier → P1/P2/P3; unmatched vertical → Rep C, flagged.

`lib/routing.ts · config/routing-rules.json`

SYMBOLIC

- ▶ zero unrouted – run fails loudly otherwise

7 Enable — battle cards

Hot/warm accounts only. 7-section card from approved proof points; Langfuse groundedness eval — an uncited claim fails the card before a rep sees it.

tools/battle-card.ts + prompts/battle-card.md · grounding corpus + blacklist/allowlist

NEURAL

→ HUMAN GATE

► battle_cards/<domain>.md – rep reviews before use (outward-facing: never graduates)

8 Decision brief

Code computes deltas vs. prior run, enrichment trust score, projections, 4Ws recommendations. The LLM optionally narrates the computed numbers — it never computes them.

tools/decision-brief.ts + memory/runs/ · 0-100-5

SYMBOLIC

NARRATION ONLY

► decision_brief.md + brief.json

02 ENRICHMENT CASCADE

Sources tried in confidence order; a fallback is never dressed as vendor data. Vendors are env-gated and probe-first — response shape verified on 1 lead before spending on 30.

Bitscale

vendor · env-gated



0.90

nRev

vendor · env-gated



0.85

DNS MX + RDAP

free · keyless · live truth



0.75

Symbolic inference

code heuristics



0.60

LLM residue

classifies · never scores



≤ 0.60

`code clamp` – the LLM fires only when every source above misses or confidence < 0.5, and its confidence is capped in code, not by prompt.

03 MEASURED NEURAL AUTHORITY – LIVE RUN

26/30

LLM-TOUCHED LEADS

Mock domains — vendors and public sources miss; rare on real data.

0/30

TIERS DECIDED BY LLM

Every hot/warm/cold call made by the rubric.

0/30

ROUTES DECIDED BY LLM

Every rep assignment made by rules + ledger.

1

WITHIN-TIER SCORE NUDGE

Only where the domain carried no evidence at all.

0–1

LLM VS. DOMAIN KEYWORD

In its only conflict with domain evidence, the LLM lost.

Coverage ≠ authority: the model touched 26 leads and decided **zero** tracked outputs.

04 CONTROL PLANE

ORCHESTRATION

Trigger.dev

TypeScript tasks · cloud-deployed · npm run pipeline needs zero keys

Per-stage retries, structured logs, shareable run link — "it runs" is a live artifact and pipeline-health instrumentation comes free.

LLM

Claude Haiku

exactly 3 touchpoints: enrichment residue · battle cards · brief narration

Cards and narration are language deliverables; everything a metric tracks stays in cheaper, testable code.

OBSERVABILITY

Langfuse

prompt · tokens · cost · latency per call

Groundedness eval per battle card; LLM spend is a dashboard number, not a surprise.

DELIVERY

Slack

production rail via Trigger.dev — not in the demo run

P1 hot-lead alert + battle card to the routed rep; run summary + threshold alerts to ops; Part 2 reuses the rail for CSM expansion alerts. Reps live in Slack — the routed lead lands where the SLA clock starts.

DECISION LOGIC

config/

icp-rubric.json · routing-rules.json · enrichment.json

Changing the ICP or territories is a config edit, never a code edit; the rubric carries its evidence citations inline.

OUTPUTS

Four artifacts

routed_leads.csv/.json · battle_cards/*.md · decision_brief.md + brief.json

CSV for the brief's deliverable, JSON for machines, cards for reps, brief for the operator.

05 DATA MODEL

Lead (source fields never mutated) → enrichment{source, confidence} → score{total, breakdown}

One Lead accretes; leads group into an Account by registrable domain. The same account shape serves both bowtie sides — Part 2 re-points enrich→score→route at the expansion book.

